IT Project Management

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Lecture 03

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- Describe the five project management process groups, the typical level of activity for each, and the interactions among them
- Relate the project management process groups to the project management knowledge areas
- Discuss how organizations develop information technology (IT) project management methodologies to meet their needs
- Review a case study of an organization applying the project management process groups to manage an IT project, describe outputs of each process group, and understand the contribution that effective initiating, planning, executing, monitoring and controlling, and closing make to project success
- Review a case study of the same project managed with an agile focus and compare the key differences between an agile approach and a predictive approach
- Describe several templates for creating documents for each process group

Project management consists of 10 knowledge areas:

1. integration,
2. scope,
3. schedule,
4. cost,
5. quality,
6. resource,
7. communications,
8. risk, procurement, and
9. stakeholder
10. management.

Projects involve **five project management process groups**

1. initiating,
2. planning,
3. executing,
4. monitoring and controlling,
5. closing

Project Life Cycle

A **project life cycle** is a collection of project phases. Projects pass through these phases from their start to their completion. The *PMBOK® Guide – Sixth Edition* describes the generic life cycle to include the following four phases:

1. Starting the project
2. Organizing and preparing
3. Carrying out the work
4. Finishing the project

These phases should not be confused with the project management process groups of initiating, planning, executing, monitoring and controlling, and closing, as described in **Chapter 3**.

In general, project life cycles define what work will be performed in each phase, what deliverables will be produced and when, who is involved in each phase, and how management will control and approve work produced in each phase. A **deliverable** is a product or service, such as a technical report, a training session, a piece of hardware, or a segment of software code, produced or provided as part of a project. (See Chapter 5, Project Scope Management, for detailed information on deliverables.)

Product Life Cycle

Just as a *project* has a life cycle, so does a *product*. IT projects help develop products and services such as new software, hardware, networks, research reports, and training on new systems. A **product life cycle** is a process used to define, create, and deliver products. Developing a product, such as a new information system, a car, a building, and so on, often involves many projects.

The PMBOK® Guide – Sixth Edition briefly describes five product or development life cycles

Two factors are important in deciding which life cycle to use:

1. the degree of change in requirements and
2. the frequency of delivery of useful results.

For example, for a product with a

low degree of change in requirements and low frequency of delivery, a predictive life cycle would be appropriate.

1. *Predictive life cycle*: The scope, schedule, and cost are determined early, and changes to scope are carefully managed. PMI also refers to predictive life cycles as waterfall.

2. *Iterative life cycle*: The scope is determined early, but time and cost estimates are modified as the understanding of the product increases. Iterations are used to develop the product through a series of repeated cycles to add to the functionality of the product. This approach works best when there is a high degree of change and a low frequency of delivery.

3. *Incremental life cycle*: Deliverables are produced through a series of iterations that add functionality within a set time frame. The deliverable is not complete until after the final iteration. This approach works best when there is a low degree of change and a high frequency of delivery.

4. *Adaptive life cycle*: Stakeholders define and approve the detailed scope before the start of an iteration, producing a useable product at the end of each iteration. PMI also refers to adaptive life cycles as agile or change-driven. This approach works best when there is a high degree of change and a high frequency of delivery.

Project Management Process Groups

A process is a series of actions directed toward a particular result.

Project managers and teams need to reexamine the business need for the project, part of monitoring and controlling activities, during every phase of the project life cycle to determine if the project is worth continuing.

- **Initiating processes** include defining and authorizing a project or project phase. Initiating processes take place during *each* phase of a project. For example, in the close-out phase, initiating processes are used to ensure that the project team completes all the work, that someone documents lessons learned, and that the customer accepts the work.

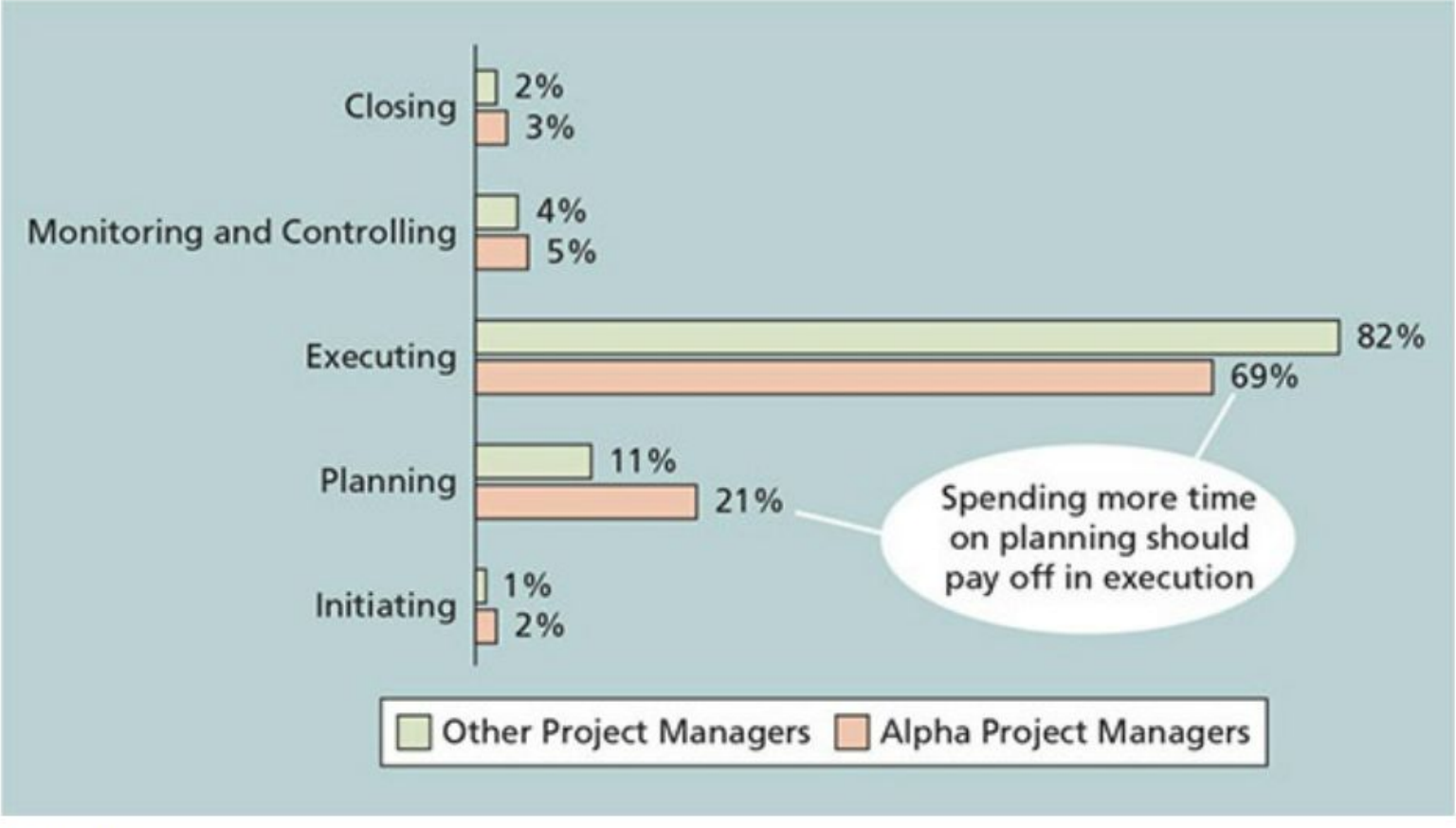
- **Planning processes** include devising and maintaining a workable scheme to ensure that the project addresses the organization's needs. Projects include several plans, such as the scope management plan, schedule management plan, cost management plan, and procurement management plan. These plans define each knowledge area as it relates to the project at a particular point in time. For example, a project team must develop a plan to define the work needed for the project, to schedule activities related to that work, to estimate costs for performing the work, and to decide what resources to procure to accomplish the work. To account for changing conditions on the project and in the organization, project teams often revise plans during each phase of the project life cycle. The project management plan, which is described in Chapter 4, coordinates and encompasses information from all other plans.

- **Executing processes** include coordinating people and other resources to carry out the various plans and create the products, services, or results of the project or phase. Examples of executing processes include directing and managing project work, managing project knowledge, acquiring resources, and conducting procurements.

- **Monitoring and controlling processes** include regularly measuring and monitoring progress to ensure that the project team meets the project objectives. The project manager and staff monitor and measure progress against the plans and take corrective action when necessary. A common monitoring and controlling process is reporting performance, where project stakeholders can identify any necessary changes that may be required to keep the project on track.

- **Closing processes** include formalizing acceptance of the project or project phase and ending it efficiently. Administrative activities are often involved in this process group, such as archiving project files, documenting lessons learned, and receiving formal acceptance of the delivered work as part of the phase or project.

Figure 3-1. Percentage of time spent on each process group



Source: Andy Crowe

Table 3-1. Project management process group and knowledge area mapping

	Project Management Process Groups				
Knowledge Area	**Initiating Process Group**	**Planning Process Group**	**Executing Process Group**	**Monitoring and Controlling Process Group**	**Closing Process Group**
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements		5.5 Validate Scope 5.6 Control Scope	

Developing an IT Project Management Methodology

Some organizations spend a great deal of time and money on training efforts for general project management skills, but after the training, project managers may still not know how to tailor their project management skills to the organization's particular needs. Because of this problem, some organizations develop their own internal IT project management methodologies. *The PMBOK® Guide* is a **standard** that describes best practices for *what* should be done to manage a project. A **methodology** describes *how* things should be done, and different organizations often have different ways of doing things.

In addition to using the *PMBOK® Guide* as a basis for project management methodology, many organizations use other guides or methods, such as the following:

- **PRojects IN Controlled Environments (PRINCE2)** : Originally developed for IT projects,

PRINCE2 was released in 1996 as a generic project management methodology by the U.K. Office of Government Commerce (OGC). It is the de facto standard in the United Kingdom and is used in over 50 countries. (See www.prince2.com for more information.) PRINCE2 defines 45 separate subprocesses and organizes them into eight process groups as follows:

1. Starting up a project
2. Planning
3. Initiating a project
4. Directing a project
5. Controlling a stage
6. Managing product delivery

Agile

RUP

SIxSIGMA

Project Pre-Initiation and Initiation

Pre-Initiation Tasks

It is good practice to lay the groundwork for a project *before* it officially starts. Senior managers often perform several tasks to lay the groundwork, sometimes called pre-initiation tasks, including the following:

- Determine the scope, time, and cost constraints for the project.
- Identify the project sponsor.

- Select the project manager.
- Develop a business case for a project.
- Meet with the project manager to review the process and expectations for managing the project.
- Determine if the project should be divided into two or more smaller projects.

Business Case to justify the investment

- Introduction/background
- Business objective
- Current situation and problem/opportunity statement
- Critical assumptions and constraints
- Analysis of options and recommendation
- Preliminary project requirements
- Budget estimate and financial analysis

- Schedule estimate
- Potential risks

Calculating NPV and ROI

7.0 Budget Estimate and Financial Analysis

A preliminary estimate of costs for the entire project is $140,000. This estimate is based on the project manager working about 20 hours per week for six months and other internal staff working a total of about 60 hours per week for six months. The customer representatives would not be paid for their assistance. A staff project manager would earn $50 per hour. The hourly rate for the other project team members would be $70 per hour, because some hours normally billed to clients may be needed for this project. The initial cost estimate also includes $10,000 for purchasing software and services from suppliers. After the project is completed, maintenance costs of $40,000 are included for each year, primarily to update the information and coordinate the Ask the Expert feature and online articles.

Projected benefits are based on a reduction in hours that consultants spend researching project management information, appropriate tools, and templates. Projected benefits are also based on a small increase in profits due to new business generated by this project. If each of 400 consultants saved just 40 hours each year (less than one hour per week) and could bill that time to projects that generate a conservative estimate of $10 per hour in profits, then the projected benefit would be $160,000 per year. If the new intranet increased business by just 1 percent, using past profit information, increased profits due to new business would be at least $40,000 each year. Total projected benefits, therefore, are about $200,000 per year. Exhibit A summarizes the projected costs and benefits and shows the estimated net present value (NPV), return on investment (ROI), and year in which payback occurs. It also lists assumptions made in performing this preliminary financial analysis. All of the financial estimates are very

encouraging. The estimated payback is within one year, as requested by the sponsor. The NPV is $272,800, and the discounted ROI based on a three-year system life is excellent at 112 percent.

Initiating

To officially initiate the project management intranet site project, Erica knew that the main tasks were to identify all of the project stakeholders and to develop the project charter. **Table 3-3** shows these processes and their outputs, based on the *PMBOK® Guide – Sixth Edition*. The main outputs are a project charter and a stakeholder register. Additional outputs that Erica found very useful for initiating projects were a stakeholder management strategy and a formal project kick-off meeting. Descriptions of how these outputs were created and sample documents related to each of them are provided for this project. Recall that every project and every organization is unique, so not all project charters, stakeholder registers, and other outputs will look the same. You will see examples of several of these documents in later chapters.

Table 3-3. Project initiation knowledge areas, processes, and outputs

Knowledge Area	Initiating Process	Outputs
Project Integration Management	Develop project charter	Project charter
		Assumption log
Project Stakeholder Management	Identify stakeholders	Stakeholder register Change requests Project management plan updates Project documents updates

Source: *PMBOK® Guide – Sixth Edition*, 2017.

Table 3-4. Stakeholder register

Name	Position	Internal/External	Project Role	Contact Information
Joe Fleming	CEO	Internal	Sponsor	joe_fleming@jwdconsulting.com
Erica Bell	PMO Director	Internal	Project Manager	erica_bell@jwdconsulting.com
Michael Chen	Senior Consultant	Internal	Team Member	michael_chen@jwdconsulting.com
Kim Phuong	Business Analyst	External	Advisor	kim_phuong@client1.com
Louise Mills	PR Director	Internal	Advisor	louise_mills@jwdconsulting.com

Table 3-5. Stakeholder management strategy

Name	Level of Interest	Level of Influence	Potential Management Strategies
Joe Fleming	High	High	Joe likes to stay on top of key projects and make money. Have a lot of short, face-to-face meetings and focus on achieving the financial benefits of the project.
Louise Mills	Low	High	Louise has a lot of things on her plate, and she does not seem excited about this project. She may be looking at other job opportunities. Show her how this project will help the company and her resume.

Table 3-6. Project charter

Project Title: Project Management Intranet Site Project

Project Start Date: May 2 **Projected Finish Date:** November 4

Budget Information: The firm has allocated $140,000 for this project. The majority of costs for this project will be internal labor. An initial estimate provides a total of 80 hours per week.

Project Manager: Erica Bell, (310) 555-5896, erica_bell@jwdconsulting.com

Project Objectives: Develop a new capability accessible on JWD Consulting's intranet site to help internal consultants and external customers manage projects more effectively. The intranet site will include several templates and tools that users can download, examples of completed templates and related project management documents used on real projects, important articles related to recent project management topics, an article retrieval service, links to other sites with useful information, and an Ask the Expert feature, where users can post questions about their projects and receive advice from experts in the field. Some parts of the intranet site will be accessible free to the public, other parts will only be accessible to current customers and internal consultants, and other parts will be accessible for a fee.

Main Project Success Criterion: The project should pay for itself within one year of completion.

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Approach:

- Develop a survey to determine critical features of the new intranet site and solicit input from consultants and customers.
- Review internal and external templates and examples of project management documents.
- Research software to provide security, manage user inputs, and facilitate the article retrieval and Ask the Expert features.
- Develop the intranet site using an iterative approach, soliciting a great deal of user feedback.
- Develop a way to measure the value of the intranet site in terms of reduced costs and new revenues, both during the project and one year after project completion.

ROLES AND RESPONSIBILITIES (PARTIAL LIST)			
Name	**Role**	**Position**	**Contact Information**
Joe Fleming	Sponsor	JWD Consulting, CEO	joe_fleming@jwdconsulting.com
Erica Bell	Project Manager	JWD Consulting, manager	erica_bell@jwdconsulting.com
Michael Chen	Team Member	JWD Consulting, senior consultant	michael_chen@jwdconsulting.com
Jessie Faue	Team Member	JWD Consulting, consultant	jessie_faue@jwdconsulting.com
Kevin Dodge	Team Member	JWD Consulting, IT department	kevin_dodge@jwdconsulting.com
Cindy Dawson	Team Member	JWD Consulting, IT department	cindy_dawson@jwdconsulting.com
Kim Phuong	Advisor	Client representative	kim_phuong@client1.com
Page Miller	Advisor	Client representative	page_miller@client2.com

Sign-Off: (Signatures of all the above stakeholders)

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

Importance of Project Phases and Management Reviews

Due to the complexity and importance of many IT projects and their resulting products, it is important to take time to review the status of a project at each phase. A project should successfully pass through each of the main project phases before continuing to the next. Because the organization usually commits more money as a project continues, a management review should occur after each phase to evaluate progress, potential success, and continued compatibility with organizational goals.

Management reviews, called **phase exits**, **phase gate reviews**, or **kill points**, are very important for keeping projects on track and determining if they should be continued, redirected, or terminated. Recall that projects are just one part of the entire system of an

Nature of IT Projects

Unlike projects in many other industries, IT projects are diverse. Some involve a small number of people installing off-the-shelf hardware and associated software. Others involve hundreds of people analyzing several organizations' business processes and then developing new software in a collaborative effort with users to meet business needs. Even for small hardware-oriented projects, a wide diversity of hardware types can be involved—personal computers, mainframe computers, network equipment, kiosks, laptops, tablets, or smartphones. The network equipment might be wireless, cellular based, or cable based, or might require a satellite connection. The nature of software development projects is even more diverse than hardware-oriented projects. A software development project might include creating a simple, stand-alone Microsoft Excel or Access application or a sophisticated, global e-commerce system that uses state-of-the-art programming languages and runs on multiple platforms.

Recent Trends Affecting Information Technology Project Management

Globalization

In his popular book *The World Is Flat*, Thomas L. Friedman describes the effects of globalization, which has created a "flat" world where everyone is connected and the "playing field" is level for many more participants.* Lower trade and political barriers and the digital revolution have made it possible to interact almost instantaneously with billions of other people across the planet, and for individuals and small companies to compete with large corporations. Friedman also discusses the increase in "uploading," in which people share information through blogging, podcasts, and open-source software.

IT is a key enabler of globalization, as shown in the following statistics. Note that total worldwide population was over 7.6 billion at the end of 2017.

Read the entire chapter 3 for next class